Quizrr Chapter-Wise Test for JEE Advanced - 2025

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Q.1

A solution of the differential equation $\left(\frac{dy}{dx}\right)^2 - \frac{dy}{dx}(e^x + e^{-x}) + 1 = 0$ is given by

(A) $y + e^{-x} = C$

(B) $y - e^{-x} = C$

(C) $y + e^{x} = C$

(D) $y - e^{2x} = C$

Q.2

If the independent variable x is changed to y, then the differential equation $x\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^3 - \frac{dy}{dx} = 0$ is transformed to $x\frac{d^2x}{dy^2} + \left(\frac{dx}{dy}\right)^2 = k$, k being a number. Then k equals

(A) 0

(B) 1

(C) -1

(D) 2

Q.3

Solution of $\sec^2 y\frac{dy}{dx} + x\tan y = x^3$ is

(A) $\tan y = x^2 + ce^{-x^2}$

(B) $\tan y = x^2 - 2 + ce^{-x^2}$

(C) $\tan y = x^2 - 2 + ce^{-x^2/2}$

(D) none of these

Q.4

The differential equation representing the family of curves $\mathrm{y}^2 = 2\mathrm{c}(\mathrm{x} + \sqrt{\mathrm{c}})$, where c is a positive parameter, is of

(A) order 1, degree 3

(B) order 1, degree 2

(C) order 2, degree 3

(D) order 2, degree 2

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Q.5

If $x^3\frac{dy}{dx} = y^3 + y^2\sqrt{y^2 - x^2}$ and $y = 1$ when $x = 1$, then $y = 2$ when $x =$

(A) $\frac{2}{5}$

(B) $\frac{4}{5}$

(C) $\frac{6}{5}$

(D) $\frac{8}{5}$

Q.6

The solutions of the differential equation $\frac{dy}{dx} = \frac{1}{xy(x^2 \sin y^2 + 1)}$ is (c is arbitrary constant)

(A) $x^2\left(\cos y^2 - \sin y^2 - 2\,\mathrm{ce}^{-y^2}\right) = 2$

(B) $y^2\left(\cos x^2 - \sin y^2 - 2e^{-y^2}\right) = 2$

(C) $x^3\left(\cos y^1 - \sin y^3 - e^{-y^2}\right) = 4\mathrm{c}$

(D) $x^2\left(\cos y^2 + \sin y^2 - 2e^{-y^2}\right) = 2$

Q.7

The curve that satisfies the differential equation, $\mathrm{y}' = \frac{\mathrm{x}^2+\mathrm{y}^2}{2\mathrm{xy}}$ and passes through $(2, 1)$ is a hyperbola with eccentricity-

(A) $\sqrt{2}$

(B) $\sqrt{3}$

(C) 2

(D) $\sqrt{5}$

Q.8

The orthogonal trajectory of the family of unit circles with centres on the x-axis which passes through the point $(0, 1)$ is

(A) $x = \sqrt{1 - y^2} - \ln\left(\frac{1+\sqrt{1-y^2}}{y}\right)$

(B) $x = -\sqrt{1 - y^2} + \ln\left(\frac{1+\sqrt{1-y^2}}{y}\right)$

(C) $x = \sqrt{1 - y^2} + \ln\left(\frac{1+\sqrt{1-y^2}}{y}\right)$

(D) $x = -\sqrt{1 - y^2} + \ln\left(\frac{1-\sqrt{1-y^2}}{y}\right)$

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Q.9

A solution of the differential equation $\left(x^2y^2-1\right)dy+2xy^3dx=0$ is -

(A) $1+x^2y^2=cx$

(B) $1+x^2y^2=cy$

(C) $\mathrm{y}=0$

(D) $y=-1/x^2$

Q.10

Solution of differential equation $\left(xy^3(1+\cos x)-y\right)dx+xdy=0$ is $\frac{x^2}{py^2}=\frac{x^3}{q}+x^2\sin x+rx\cos x-2\sin x+c$. Find the value less than $p+q+r$.

(A) 2

(B) 3

(C) 5

(D) 7

Q.11

If $\mathrm{f}(\mathrm{x})$ is a function such that $\int_0^x(1-t)f(t)dt=\int_0^x tf(t)dt$ and $\mathrm{f}(1)=1$ then find $f(x)$

(A) $e^{-1/x-3\log x-1}$

(B) $e^{-\frac{1}{x}-3\log x+1}$

(C) $\frac{1}{x^3}e^{-\frac{1}{x}-1}$

(D) None of these

Q.12

Passage

A differential equation of the form $\frac{d^2y}{dx^2}+a_1\frac{dy}{dx}+a_2y=0$ is called linear differential equation where a_1 and a_2 are function of x only. In case a_1 and a_2 are constants, the solution of the linear differential equation can be easily written by following facts-

(i) $\mathrm{y}=0$ is a solution of differential equation

(ii) If $y=f(x)$ is a solution then $y=cf(x)$ is also a solution.

(iii) If $y=f_1(x)\&y=f_2(x)$ are two solution then $y=f_1(x)+f_2(x)$ will also be a solution.

(iv) If distinct roots of quadratic equation $m^2+a_1m+a_2=0$ are m_1 and m_2 (real or imaginary) then solution of differential equation is $y=c_1e^{m_1x}+c_2e^{m_2x}$.

In case roots are complex the solution can be transformed to form $\mathrm{e}^{\mathrm{ax}}\left(\mathrm{c}_1\cos\mathrm{bx}+\mathrm{c}_2\sin\mathrm{bx}\right)$ by using Euler's theorem.

(v) In case roots of $m^2+a_1m+a_2=0$ are equal $\left(\text{ say } m_1\right)$ the differential equation can be make linear by putting $\frac{dy}{dx}-m_1y=v$.

The linear differential equation can also be satisfied by some other functions which are not of the above type such

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functions are called particular integrals.

Question:

$y = e^x$ is a solution of differential equation $\frac{d^2y}{dx^2} - y = 0$, then which of the following is not a solution.

(A) e^{-x}

(B) $e - x$

(C) $ae^x + be^{-x}$

(D) $e^x + c$

Q.13

Passage

A differential equation of the form $\frac{d^2y}{dx^2} + a_1\frac{dy}{dx} + a_2y = 0$ is called linear differential equation where a_1 and a_2 are function of x only. In case a_1 and a_2 are constants, the solution of the linear differential equation can be easily written by following facts-

(i) $y = 0$ is a solution of differential equation

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(iv) If distinct roots of quadratic equation $m^2 + a_1m + a_2 = 0$ are m_1 and m_2 (real or imaginary) then solution of differential equation is $y = c_1e^{m_1x} + c_2e^{m_2x}$.

In case roots are complex the solution can be transformed to form $e^{ax}\left(c_1\cos bx + c_2\sin bx\right)$ by using Euler's theorem.

(v) In case roots of $m^2 + a_1m + a_2 = 0$ are equal $\left(\text{ say } m_1\right)$ the differential equation can be make linear by putting $\frac{dy}{dx} - m_1y = v$.

The linear differential equation can also be satisfied by some other functions which are not of the above type such functions are called particular integrals.

Question:

Which of the following is solution of the equation $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 9y = 0$

(A) $c_1 + c_2x$

(B) $\left(c_1 + c_2x\right)e^x$

(C) $c_1\cos 3x + c_2\sin 3x$

(D) None of these

Q.14

Passage

A differential equation of the form $\frac{d^2y}{dx^2} + a_1\frac{dy}{dx} + a_2y = 0$ is called linear differential equation where a_1 and a_2 are function of x only. In case a_1 and a_2 are constants, the solution of the linear differential equation can be easily written by following facts-

(i) $y = 0$ is a solution of differential equation

(ii) If $y = f(x)$ is a solution then $y = cf(x)$ is also a solution.

(iii) If $y = f_1(x) \& y = f_2(x)$ are two solution then $y = f_1(x) + f_2(x)$ will also be a solution.

(iv) If distinct roots of quadratic equation $m^2 + a_1m + a_2 = 0$ are m_1 and m_2 (real or imaginary) then solution of differential equation is $y = c_1e^{m_1x} + c_2e^{m_2x}$.

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In case roots are complex the solution can be transformed to form $e^{ax}\left(c_1 \cos bx + c_2 \sin bx\right)$ by using Euler's theorem.

(v) In case roots of $m^2 + a_1 m + a_2 = 0$ are equal (say m_1) the differential equation can be make linear by putting $\frac{dy}{dx} - m_1 y = v$.

The linear differential equation can also be satisfied by some other functions which are not of the above type such functions are called particular integrals.

Question:

A particular integral solution of the equation $\frac{d^2y}{dx^2} + a^2 y = \sin ax$ is -

(A) $c_1 \cos(ax) + c_2 \sin(ax)$

(B) $\frac{x \sin ax}{2a^2} + \frac{x \cos ax}{2a^2}$

(C) $\frac{x \sin ax}{2a^2} + \frac{\cos ax}{2a^2}$

(D) $\frac{-x \sin ax}{2a^2} - \frac{\cos ax}{2a^2}$

Q.15

Passage

Solution to Differential equation can be used to solve large number of practical problems like growth decay problems, Newton Cooling law, Dilution problems, Compound interest problems and large number of geometrical problems.

Let $\mathrm{N(t)}$ denote the amount of substance (or population) that is either growing or decaying. If we assume that $\mathrm{dN/dt}$, the time rate of change of this amount of substance, is proportional to the amount of substance present, then For growth : $\frac{\mathrm{dN}}{\mathrm{dt}} = \mathrm{kN}$, or $\frac{\mathrm{dN}}{\mathrm{dt}} - \mathrm{kN} = 0$, For decay $\frac{\mathrm{dN}}{\mathrm{dt}} = -\mathrm{kN}$

Where k is the constant of proportionality. We are assuming that $\mathrm{N(t)}$ is a differentiable, hence continuous, function of time.

Question:

A tank contains 30 lit. of a chemical solution prepared by dissolving $120\mathrm{gm}$ of a soluble substance in the fresh water. Fluid containing $4\mathrm{gm}$. of this substance per lit. runs in at the rate of $4\mathrm{lit}$./min. and the well-stirred mixture runs out at the same rate. The amount of substance in the tank after 30 min. is

(A) $0\mathrm{gm}$.

(B) $120\mathrm{gm}$.

(C) $60\mathrm{gm}$.

(D) $80\mathrm{gm}$.

Q.16

Passage

Solution to Differential equation can be used to solve large number of practical problems like growth decay problems, Newton Cooling law, Dilution problems, Compound interest problems and large number of geometrical problems.

Let $N(t)$ denote the amount of substance (or population) that is either growing or decaying. If we assume that $\mathrm{dN/dt}$, the time rate of change of this amount of substance, is proportional to the amount of substance present, then For growth : $\frac{\mathrm{dN}}{\mathrm{dt}} = \mathrm{kN}$, or $\frac{\mathrm{dN}}{\mathrm{dt}} - \mathrm{kN} = 0$, For decay $\frac{\mathrm{dN}}{\mathrm{dt}} = -\mathrm{kN}$

Where k is the constant of proportionality. We are assuming that $\mathrm{N(t)}$ is a differentiable, hence continuous, function

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of time.

Question:

A population grows at the rate of 5% per year. Then the population will be doubled in -

(A) $10 \log 2$ years

(B) $20 \log 2$ years

(C) $30 \log 2$ years

(D) None of these

Q.17

Passage

Solution to Differential equation can be used to solve large number of practical problems like growth decay problems, Newton Cooling law, Dilution problems, Compound interest problems and large number of geometrical problems.

Let $N(t)$ denote the amount of substance (or population) that is either growing or decaying. If we assume that dN/dt, the time rate of change of this amount of substance, is proportional to the amount of substance present, then

For growth : $\frac{\mathrm{dN}}{\mathrm{dt}} = \mathrm{kN}$, or $\frac{\mathrm{dN}}{\mathrm{dt}} - \mathrm{kN} = 0$, For decay $\frac{\mathrm{dN}}{\mathrm{dt}} = -\mathrm{kN}$

Where k is the constant of proportionality. We are assuming that $\mathrm{N(t)}$ is a differentiable, hence continuous, function of time.

Question:

Spherical rain drop evaporates at a rate proportional to its surface area. The differential equation corresponding to the rate of change of the radius of the rain drop if the constant of proportionality is $\mathrm{K} > 0$, is

(A) $\frac{\mathrm{dr}}{\mathrm{dt}} + \mathrm{K} = 0$

(B) $\frac{\mathrm{dr}}{\mathrm{dt}} - \mathrm{K} = 0$

(C) $\frac{\mathrm{dr}}{\mathrm{dt}} = \mathrm{Kr}$

(D) none

Q.18

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Match the column -

(A) The degree of the differential equation, of which $y^2 = 4a(x + a)$ is a solution, is – (p) 2

(B) The degree of differential equation (q) 1

$$x = 1 + \left(\frac{dy}{dx}\right) + \frac{1}{2!}\left(\frac{dy}{dx}\right)^2 + \frac{1}{3!}\left(\frac{dy}{dx}\right)^3 + \ldots\ldots$$

(C) The degree of the differential equation (r) 3

$$\left(\frac{d^4y}{dx^4}\right)^{3/5} - 5\frac{d^3y}{dx^3} + 6\frac{d^2y}{dx^2} - 8\frac{dy}{dx} + 5 = 0 \text{ is} -$$

(D) The order of the differential equation associated with the primitive $y = c_1 + c_2e^x + c_3e^{-2x+c_4}$, where c_1, c_2, c_3, c_4 are arbitrary constants, is (s) 4 (t) 5

(A) A-q ; B-r ; C-p ; D-q

(B) A-p ; B-q ; C-s ; D-t

(C) A-p ; B-q ; C-r ; D-r

(D) A-p ; B-s ; C-t ; D-r

Q.19

An inverted conical tank of 2 m radius and 4 meter height to initially full of water, has an outlet at bottom. The outlet is opened at some instant. The rate of flow through the outlet at any time t is $6h^{3/2}$, where h is height of water level above the outlet at time t. Find the sum of α and β if $t = \frac{\alpha\pi}{\beta}$ (where α and β are in least $(+)$ integer forms).

Q.20

If the differential equation of a curve, passing through $\left(0, -\frac{\pi}{4}\right)$ and $(t, 0)$ is $\cos y\left(\frac{dy}{dx} + e^{-x}\right) + \sin y\left(e^{-x} - \frac{dy}{dx}\right) = e^{e^{-x}}$, then find the value of $t \cdot e^{e^{-t}}$.

Answers & Solutions

Q.1 Answer:

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Solution:

The given equation can be written as

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$\left(\frac{dy}{dx}-e^{-x}\right)\left(\frac{dy}{dx}-e^{x}\right)=0$

$\Rightarrow \frac{dy}{dx}-e^{-x}=0 \,\&\, \frac{dy}{dx}-e^{x}=0$

$\Rightarrow\ dy-e^{-x}\,dx=0 \,\&\, dy-e^{x}\,dx=0$

$\Rightarrow\ \ y+e^{-x}=C \,\&\,\ y-e^{x}=C.$

Q.2 Answer:

1

Solution:

$\frac{dy}{dx}=\frac{1}{dx/dy}, \frac{d^2y}{dx^2}=-\frac{-1}{(dx/dy)^3}\frac{d^2x}{dy^2}$
then $x\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^3-\left(\frac{dy}{dx}\right)=0$ **changes to**
$x\frac{d^2x}{dy^2}+\left(\frac{dx}{dy}\right)^2-1=0$

Q.3 Answer:

$\tan y=x^2-2+ce^{-x^2/2}$

Solution:

Put $\tan y=z$
$\therefore \sec^2 y\frac{dy}{dx}=\frac{dz}{dx}$
$\therefore \frac{dz}{dx}+xz=x^3$ **which is linear in** z
$\therefore \quad \int Pdx=\int xdx=\frac{x^2}{2} \quad \therefore e^{\int Pdx}=e^{x^2/2}$
$\therefore \quad$ solution is $ze^{x^2/2}=\int e^{x^2/2}\cdot x^3dx=\int e^{x^2/2}\cdot x^2\cdot xdx$
Put $\quad \frac{x^2}{2}=t$
$\therefore xdx=dt$
$=\int e^t\cdot 2tdt=2e^t(t-1)+c$
$\Rightarrow \quad \tan y=2\left(\frac{x^2}{2}-1\right)+ce^{-x^2/2}=x^2-2+ce^{-x^2/2}$

Q.4 Answer:

order 1, degree 3

Solution:

$y^2=2c(x+\sqrt{c})\Rightarrow 2yy_1=2c\Rightarrow yy_1=c$
eliminating 'c'
$y^2=2yy_1\left(x+\sqrt{yy_1}\right)$
$\Rightarrow y-2xy_1=2\sqrt{y}\cdot y_1^{3/2}$
$(y-2xy_1)^2=4yy_1^3$
It is a differential equation of order 1 and degree 3.

Q.5 Answer:

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$\frac{8}{5}$

Solution:

It is homogeneous d.e $y = vx \Rightarrow \frac{xdv}{dx} = v^3 + v^2\sqrt{v^2-1} - v$

$\frac{dx}{x} = \frac{dv}{v\sqrt{v^2-1}\left(v+\sqrt{v^2-1}\right)} = \frac{v-\sqrt{v^2-1}}{v\sqrt{v^2-1}}dv = \left(\frac{1}{\sqrt{v^2-1}} - \frac{1}{v}\right)dv$

Integrating $cx = \frac{v+\sqrt{v^2-1}}{v} = \frac{y+\sqrt{y^2-x^2}}{y}$

$x = 1, y = 1 \Rightarrow c = 1$

$y = 2 \Rightarrow 2(x-1) = \sqrt{4-x^2} \Rightarrow (5x-8)x = 0 \Rightarrow x = \frac{8}{5}$

Q.6 Answer:

$x^2\left(\cos y^2 - \sin y^2 - 2\,\mathrm{ce}^{-y^2}\right) = 2$

Solution:

$$\frac{dx}{dy} = xy\left(x^2\sin y^2 + 1\right)$$

$$\frac{1}{x^3}\frac{dx}{dy} - \frac{1}{x^2}y = y\sin y^2$$

Put $\frac{-1}{x^2} = u$

$\frac{dx}{dy} + 2uy = 2y\sin y^2$ **which is linear equation.**

$$ue^{y^2} = \int 2y\sin y^2 e^{y^2}dy + C$$

$$= \frac{1}{2}e^{y^2}\left(\sin y^2 - \cos y^2\right) + C$$

$$\Rightarrow x^2\left(\cos y^2 - \sin y^2 - 2ce^{-y^2}\right) + c$$

Q.7 Answer:

$\sqrt{2}$

Solution:

With $y = xv, xv' = \frac{1+v^2}{2v} - v = \frac{1-v^2}{2v}$

$\Rightarrow \frac{2\mathrm{v}d\mathrm{v}}{1-\mathrm{v}^2} = \frac{d\mathrm{x}}{\mathrm{x}}$ **whose solution is** $\mathrm{x}\left(1-\mathrm{v}^2\right) = \mathrm{c}$.

When $\mathrm{x} = 2, \mathrm{v} = \frac{1}{2}$ **so that** $\mathrm{c} = \frac{3}{2}$.

The equation of the curve is $x^2 - y^2 = \frac{3x}{2} \Rightarrow \left(x - \frac{3}{4}\right)^2 - y^2 = \frac{9}{16}$.

This is a rectangular hyperbola with eccentricity $\sqrt{2}$.

Q.8 Answer:

$x = -\sqrt{1-y^2} + \ln\left(\frac{1+\sqrt{1-y^2}}{y}\right)$

Solution:

The circles are $(x-h)^2 + y^2 = 1$

Differentiating, $x - h + yy_1 = 0$

Eliminating $x - h, y_1^2 = \frac{1-y^2}{y^2} \Rightarrow y_1 = \frac{\sqrt{1-y^2}}{y}$

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Replacing y_1 by $-\frac{1}{y_1}, y_1 = -\frac{y}{\sqrt{1-y^2}} \Rightarrow \frac{\sqrt{1-y^2}}{y}dy + dx = 0$

$\Rightarrow x + c = -\int \frac{\sqrt{1-y^2}}{y}dy, y = \sin\theta$

$= -\int \frac{\cos^2\theta}{\sin\theta}d\theta = \int(\sin\theta - \operatorname{cosec}\theta)d\theta$

$= -\cos\theta + \ln(\operatorname{cosec}\theta + \cot\theta)$

$x = 0, y = 1, \theta = \frac{\pi}{2} \Rightarrow c = 0$

$x = -\cos\theta + \ln\left(\frac{1+\cos\theta}{\sin\theta}\right) = -\sqrt{1-y^2} + \ln\left(\frac{1+\sqrt{1-y^2}}{y}\right)$

Q.9 Answer:

$1 + x^2y^2 = cy$

Solution:

$\left(x^2y^2 - 1\right)dy + 2xy^3dx = 0$

$x^2y^2dy + 2xy^3dx = dy$

$x^2dy + 2xydx = \frac{dy}{y^2}; \int d\left(x^2y\right) = \int \frac{dy}{y^2} + c$

$\Rightarrow x^2y = \frac{y^{-1}}{-1} + c \Rightarrow x^2y^2 = -1 + cy \Rightarrow 1 + x^2y^2 = cy$

Q.10 Answer:

2

Solution:

The given equation can be rewritten as

$\frac{dy}{dx} + y^3(1+\cos x) - \frac{y}{x} = 0$ **i.e.,** $\frac{1}{y^3}\frac{dy}{dx} - \frac{1}{y^2x} = -(1+\cos x)$

Using the transformation $-\frac{1}{y^2} = u.$

we get $\frac{2}{y^3}dy = du$

The above equation reduces to $\frac{1}{2}\frac{du}{dx} + \frac{u}{x} = -(1+\cos x)$

whose integrating factor $= e^{\int \frac{2}{x}dx} = e^{2\ln x} = x^2$

Hence, the solution of the given differential equation is given by $ux^2 = -2\int x^2(1+\cos x)dx$

i.e., $\frac{x^2}{2y^2} = \int x^2dx + \int x^2\cos xdx$

$\frac{x^2}{2y^2} = \frac{x^3}{3} + x^2\sin x - \int 2x\sin xdx$

$\frac{x^2}{2y^2} = \frac{x^3}{3} + x^2\sin x + 2x\cos x - 2\int\cos x + c$

$\frac{x^2}{2y^2} = \frac{x^3}{3} + x^2\sin x + 2x\cos x - 2\sin x + c$

Q.11 Answer:

$e^{-1/x - 3\log x - 1}$

Solution:

$x\int_0^x(1-t)f(t)dt = \int_0^x tf(t)dt$ **...(a)**

Differentiate both side eq. (a)

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$$\int_0^x (1-t)f(t)dt + x(1-x)f(x) = xf(x)$$

$$\int_0^x (1-t)f(t)dt = x^2 f(x) \quad \ldots(b)$$

Again differentiate eq. (b) both side

$$(1-x)f(t) = 2xf(x) + x^2 f'(x)$$

$$(1-3x)f(x) = x^2 f'(x)$$

$$\int \frac{1-3x}{x^2}dx = \int \frac{f'(x)}{f(x)}dx + c$$

$$\int \left(\frac{1}{x^2} - \frac{3}{x}\right)dx = \int \frac{f'(x)}{f(x)} + c$$

$$-\frac{1}{x} - 3\log x = \log_e f(x) + c$$

$$c = -1$$

$$\log_e f(x) = -\frac{1}{x} - 3\log x - 1$$

$$f(x) = e^{-1/x - 3\log x - 1}$$

Q.12 Answer:

$e - x$

Solution:

(i) Obviously option (B) is not a solution.

(ii) Among the four option (B) is the solution of the differential equation

(iii) Among the four option (D) is satisfying.

Q.13 Answer:

None of these

Solution:

(i) Obviously option (B) is not a solution.

(ii) Among the four option (B) is the solution of the differential equation

(iii) Among the four option (D) is satisfying.

Q.14 Answer:

$\frac{-x\sin ax}{2a^2} - \frac{\cos ax}{2a^2}$

Solution:

(i) Obviously option (B) is not a solution.

(ii) A mong the four option (B) is the solution of the differential equation

(iii) Among the four option (D) is satisfying.

Q.15 Answer:

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120gm.

Solution:

As concentration of input and that of tank is same (also rate of in flow = rate of out flow), then the amount of substance in the tank will remain same.

Q.16 Answer:

$20 \log 2$ years

Solution:

Let the population be p at any time t.

Then $\frac{dp}{dt} = 5\%$ **of** $p \Rightarrow \frac{dp}{dt} = \frac{p}{20} \Rightarrow \frac{dp}{p} = \frac{dt}{20}$

Integrating, we get $\log p = \frac{t}{20} + c$

Let the population be $\mathrm{p_0}$ **at** $\mathrm{t} = 0$

Putting $p = p_0$ **and** $t = 0$ **in** (1)**, we get** $c = \log p_0$**.**

$\therefore \quad \log p = \frac{t}{20} + \log p_0$

Let $\mathrm{p} = 2\mathrm{p_0}$ **at** $\mathrm{t} = \mathrm{t_1}$ **Then,** $\log(2\mathrm{p_0}) = \frac{\mathrm{t_1}}{20} + \log \mathrm{p_0}$

$\Rightarrow \log 2 + \log \mathrm{p_0} = \frac{\mathrm{t_1}}{20} + \log \mathrm{p_0}$

$\Rightarrow \log 2 = \frac{t_1}{20} \Rightarrow t_1 = 20(\log 2)$**, which is the required time.**

Q.17 Answer:

$\frac{\mathrm{dr}}{\mathrm{dt}} + \mathrm{K} = 0$

Solution:

$\frac{\mathrm{dV}}{\mathrm{dt}} = -\mathrm{k}4\pi\mathrm{r}^2$

but $V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$

hence $\frac{dr}{dt} = -K$

Q.18 Answer:

A-p ; B-q ; C-r ; D-r

Solution:

(A) – p, (B) – q, (C) – r, (D) – r

(A) We have, $y^2 = 4a(x + a)$

Differentiating w.r.t. x**, we get** $2y\frac{dy}{dx} = 4$ **a i.e.,** $a = \frac{y}{2}\frac{dy}{dx}$

Substituting the value of a in (1), we get $y^2 = 2y\frac{dy}{dx}\left[x + \frac{y}{2}\frac{dy}{dx}\right]$

or $y = 2x\frac{dy}{dx} + y\left(\frac{dy}{dx}\right)^2$ **or** $y\left[1 - \left(\frac{dy}{dx}\right)^2\right] = 2x\frac{dy}{dx}$

which is the required differential equation.

The degree of the differential equation is 2.

(B) $\mathrm{x} = \mathrm{e}^{(\mathrm{dy/dx})} \Rightarrow \frac{\mathrm{dy}}{\mathrm{dx}} = \ln \mathrm{x} \quad \therefore$ **order** $= 1$ **and degree** $= 1$

(C) We have, $\left(\frac{d^4y}{dx^4}\right)^{3/5} = 5\frac{d^3y}{dx^3} - 6\frac{d^2y}{dx^2} + 8\frac{dy}{dx} - 5$

$\left(\frac{d^4y}{dx^4}\right)^3 = \left(5\frac{d^3y}{dx^3} - 6\frac{d^2y}{dx^2} + 8\frac{dy}{dx} - 5\right)^5$

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which is a differential equation of order 4 and degree 3.

(D) $y = C_1 + C_2e^x + C_3e^{-2x+c4} = C_1 + C_2e^x + C_3e^{c4} \cdot e^{-2x}$

So, Differential equation will be order of 3.

Q.19 Answer:

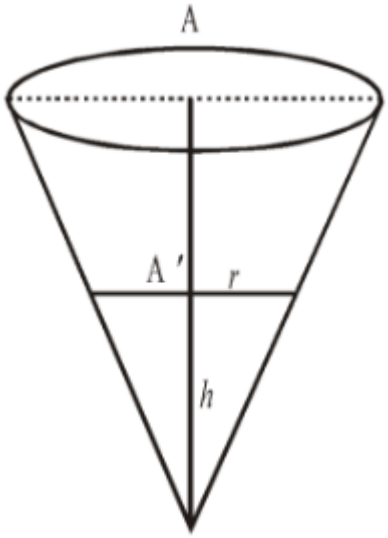
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Solution:

Given that $\frac{dv}{dt} = 6h^{3/2}$

at time ' t **' the volume of water in the tank is** $v = 1/3\pi r^2 h$

$\frac{OA'}{A'B'} = \frac{OA}{AB} \Rightarrow \frac{h}{r} = \frac{4}{2} \Rightarrow r = \frac{h}{2}$



$$V = \frac{1}{3}\pi\frac{h^3}{4} \Rightarrow h^3 = \frac{12V}{\pi}$$

$$\Rightarrow \frac{dV}{dt} = \frac{-6\sqrt{12}}{\sqrt{\pi}} \times \sqrt{V}$$

$$\int_{v_0}^{0} \frac{dV}{\sqrt{V}} = \frac{-6\sqrt{12}}{\sqrt{\pi}} \int_0^t dt \Rightarrow (2\sqrt{V})_{v_0}^0 = \frac{-6\sqrt{12}}{\pi} = t$$

$$t = \frac{2\sqrt{V_0}\sqrt{\pi}}{6\sqrt{12}} = \frac{2\sqrt{\pi}}{2\sqrt{12}} \times \sqrt{\frac{16}{3}}\pi \Rightarrow t = \frac{2\pi}{9} \text{ unit}$$

$$\alpha + \beta = 11$$

Q.20 Answer:

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Solution:

$$(\cos y - \sin y)\frac{dy}{dx} + (\cos y + \sin y)e^{-x} = e^{e^{-x}}$$

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$$\cos y + \sin y = u$$

$$\Rightarrow \frac{du}{dx} + e^{-x}u = e^{-x} \Rightarrow ue^{-e^{-x}} = x$$

$$\Rightarrow e^{-e^{e^{-t}}} = t \text{ (Putting } (t, 0) \text{)}$$

$$\Rightarrow \quad t \cdot e^{e^{-t}} = 1$$

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